

Supply Chain Cost Forecasting and Data Analytics in the Automobile Manufacturing Industry

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Keywords:

Supply Chain Management
Automobile Industry
Cost Forecasting
Data Analytics
Power BI
Predictive Analytics

ABSTRACT

The automobile manufacturing industry operates through highly complex and globally dispersed supply chains that are increasingly exposed to uncertainty, cost pressure, and operational disruptions. Traditional supply chain management approaches, primarily focused on cost minimization and efficiency, are no longer adequate to address volatility in demand, globalization-related risks, and rising customer expectations. This research paper investigates the application of data analytics and forecasting techniques for supply chain cost management in the automobile manufacturing sector. Using historical supply chain cost data of 100 weeks, forecasting models are developed and implemented using Power BI to analyze and predict key cost components such as child part cost, equipment cost, manpower cost, tool maintenance cost, transportation cost, and total supply chain cost. The results demonstrate that data-driven forecasting enables improved cost visibility, proactive decision-making, and enhanced profitability. The study confirms that Power BI serves as an effective business intelligence tool for supply chain forecasting and cost optimization in automobile manufacturing.

Journal link:

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1. Introduction

Globalization has significantly intensified over recent decades, leading to supply chains that are wider, more complex, and geographically dispersed across multiple countries and cultures [1]. The automobile manufacturing industry exemplifies this complexity, as it involves extensive networks of raw material suppliers, component manufacturers, original equipment manufacturers, logistics providers, and distributors operating under diverse economic and regulatory conditions [2]. At the

same time, managers face constant pressure to improve efficiency by reducing costs, shortening lead times, minimizing inventory, and adopting just-in-time and lean manufacturing practices [3]. While such efficiency-driven strategies improve competitiveness under stable conditions, they also increase vulnerability to supply chain risks [4]. Late deliveries, inaccurate demand forecasting, supplier failures, natural disasters, and geopolitical disruptions can interrupt material flow and planned operations. Because modern supply chains are highly interconnected, disruptions at one node can propagate rapidly across the entire network. Consequently, organizations have recognized the need to manage supply chains as integrated systems rather than isolated organizational units [5].

Supply Chain Management (SCM) addresses this requirement by coordinating material, information, and financial flows from suppliers' suppliers to customers' customers in order to enhance customer value and achieve competitive advantage. In recent years, data analytics has emerged as a critical enabler for effective SCM, particularly in forecasting demand and costs, improving visibility, and supporting strategic and operational decision-making. This research focuses on applying data analytics and forecasting techniques to supply chain cost management in the automobile manufacturing industry [6].

1.1 Supply Chain Management and Supply Chain Cost

Supply Chain Management is concerned with the efficient integration of suppliers, manufacturers, warehouses, and distribution channels to ensure that the right product is delivered at the right time, location, and cost. Traditional definitions of SCM emphasize cost minimization, customer service, inventory management, and material flow coordination. The classical objective of SCM is to minimize total supply chain cost while meeting given demand levels [7].

Supply chain cost includes raw material acquisition costs, inbound transportation costs, facility investment costs, direct and indirect manufacturing costs, inventory holding costs, inter-facility transportation costs, and outbound distribution costs. In modern supply chains, a large portion of these costs lies outside the legal boundaries of the focal firm due to extensive outsourcing of production and logistics activities. Therefore, a supply chain-wide cost perspective is essential for effective cost management and optimization [8].

1. 2 Supply Chain Challenges in the Automobile Industry

The automobile industry faces several persistent supply chain challenges. Limited visibility across multi-tier supplier networks restricts effective planning and risk mitigation. Cost containment remains difficult due to fluctuating raw material prices, labor costs, and energy expenses. Customer expectations for quality, customization, and timely delivery continue to rise, adding complexity to supply chain coordination [9-13].

In the Indian automotive context, these challenges are further intensified by fragmented supplier bases, infrastructure limitations, complex tariffs, and a multilevel distribution system. The auto component industry, a key contributor to industrial growth and exports, is particularly affected by frequent changes in raw material costs and strict delivery schedules. Integration of end-to-end supply chains remains one of the most significant challenges for automotive manufacturers [14-16].

1.3 Supply Chain Risk Management and Disruptions

Supply chain risk management involves identifying, assessing, and mitigating risks across the entire supply chain. Risks may originate from external sources such as demand volatility, supplier failures, environmental events, geopolitical instability, and regulatory changes, or from internal sources such as manufacturing breakdowns, poor planning, and inadequate contingency measures. Supply chain disruptions are events that interrupt the normal flow of materials, information, or finances. The COVID-19 pandemic highlighted that cost optimization alone is insufficient, as the cost of reacting to disruptions often exceeds the cost of preparedness. Effective disruption management requires rapid assessment of critical events, evaluation of supplier viability, securing supply continuity, and ensuring customer commitments are met.

1.4 Role of Data Analytics in Supply Chain Management

Data analytics refers to the systematic examination of data to extract insights that support managerial decision-making. In supply chain management, analytics enables organizations to process large volumes of operational data that exceed human cognitive capabilities. By leveraging data analytics, firms can improve forecasting accuracy, enhance operational efficiency, reduce costs, and gain competitive advantage.

Supply chain analytics typically includes descriptive analytics to understand historical performance, predictive analytics to forecast future outcomes, and prescriptive analytics to recommend optimal decisions. Forecasting accuracy is particularly critical in automotive supply chains due to long lead times, high capital investment, and the risk of product obsolescence

1.5 Research Gap and Objectives

Despite extensive literature on supply chain management and big data analytics, empirical research focusing on cost forecasting using real industrial data in the automobile manufacturing sector remains limited. Moreover, practical applications of low-code analytics platforms such as Power BI have received insufficient academic attention.

The objectives of this research are:

1. To analyze historical supply chain cost data in the automobile manufacturing industry.
2. To apply forecasting techniques using Power BI for key cost components.
3. To evaluate the impact of cost forecasting on supply chain decision-making and profitability.

2. Research Methodology

This study adopts an empirical data analytics approach. Historical supply chain cost data covering 100 weeks were collected from an automobile parts manufacturing context. The dataset includes child part cost, equipment cost, manpower cost, tool maintenance cost, transportation cost, miscellaneous cost, and total supply chain cost.

Power BI was used as the primary analytical tool due to its capability for data integration, visualization, and forecasting without extensive coding. Time-series forecasting was applied to predict future cost trends, and confidence intervals were generated to assess forecasting reliability. The methodology emphasizes practical applicability and managerial relevance

2.1 Overview of Power BI

Power BI is a cloud-based business intelligence platform that allows users to connect, transform, analyze, and visualize data from multiple sources. It consists of Power BI Desktop for data modeling and report creation, Power BI Service for publishing and collaboration, and Power BI Mobile for real-time access. Its forecasting feature supports time-series analysis and visualization, making it suitable for supply chain cost analysis and decision support.

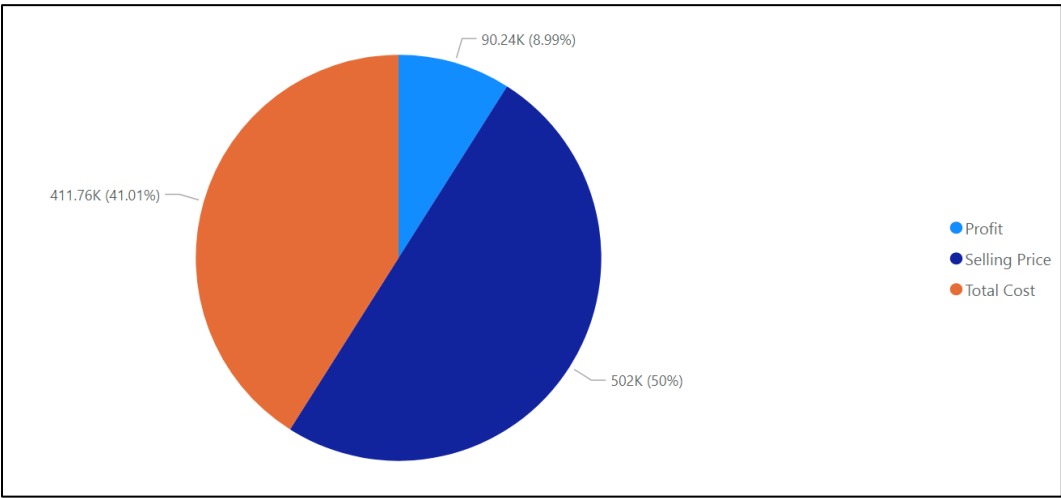
3 Analysis of Supply Chain Cost, Profit, and Selling Price

3.1 Comparison of Profit, Selling Price, and Total Cost

Analytics represents the ability of organizations to make data-driven decisions by transforming raw operational data into meaningful insights, often through visualization techniques such as graphs and dashboards. In the context of supply chain management, analytics refers to the systematic analysis of historical and real-time data generated across supply chain processes in order to enhance planning accuracy, operational efficiency, and managerial decision-making.

In this study, a comparative analysis of profit, selling price, and total supply chain cost was carried out using a pie chart representation (Graph 8). The visualization summarizes the proportional contribution of each component to the overall financial structure. The results indicate that the selling price accounts for approximately 50% of the total financial value, while total supply chain cost constitutes 41.01%, and profit contributes only 8.99%.

This distribution highlights a relatively narrow profit margin in the automobile manufacturing supply chain, emphasizing the critical importance of cost control and efficiency improvement. The dominance of total cost within the selling price structure suggests that even marginal reductions in supply chain cost can lead to significant improvements in profitability. Therefore, analytics-driven cost monitoring and forecasting play a vital role in identifying cost-saving opportunities and strengthening financial sustainability.



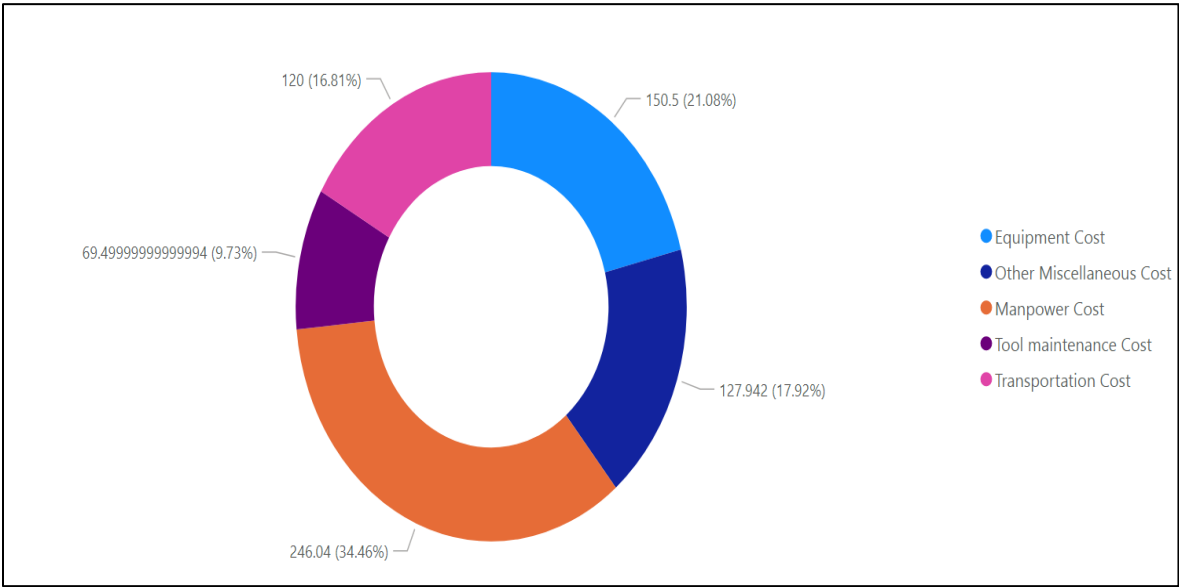
Graph 1 Profit, Selling Price, Total Cost comparison

3.2 Cost Bifurcation Analysis

To gain deeper insight into the composition of total supply chain cost, a cost bifurcation analysis was performed using a donut chart (Graph 9). This visualization illustrates the relative contribution of individual cost components to the overall supply chain cost structure.

The analysis reveals that manpower cost is the largest contributor, accounting for 34.46% of the total cost. This indicates a high dependency on labor-intensive operations within the supply chain and highlights manpower cost as a critical target for optimization. Equipment cost emerges as the second-largest contributor at 21.08%, followed by miscellaneous costs at 17.92%, transportation cost at 16.81%, and tool maintenance cost at 9.73%.

The dominance of manpower cost suggests that productivity enhancement, workforce optimization, and automation initiatives could substantially reduce total supply chain cost. Similarly, the combined contribution of equipment, transportation, and maintenance costs reflects the capital-intensive and logistics-driven nature of automobile manufacturing. Cost bifurcation analysis therefore provides valuable decision support by clearly identifying cost drivers and enabling managers to prioritize improvement initiatives based on their financial impact.



Graph 2 Cost wise Bifurcation

3.3 Month-wise Analysis of Profit, Cost, and Selling Price

A month-wise comparative analysis of profit, total cost, and selling price was conducted using a clustered column chart (Graph 10). This visualization enables the examination of temporal variations and facilitates the identification of periods with relatively high or low profitability. The analysis reveals noticeable fluctuations across months, reflecting changes in market demand, operational efficiency, and cost behavior. The total supply chain cost shows a declining trend starting in December 2022, with a reduction of approximately 0.12% (20.31 units) over a two-month period. This decline indicates the effectiveness of cost control measures and operational adjustments implemented during this phase.

Conversely, profit exhibits a significant drop between May 2021 and February 2022, decreasing from 4,976.85 to 3,567.95 during its steepest decline. This reduction may be attributed to demand volatility, rising input costs, or inefficiencies within supply chain operations during that period. The divergence between selling price stability and profit volatility further reinforces the sensitivity of profitability to cost fluctuations.

Overall, the month-wise analysis demonstrates the importance of continuous monitoring of financial performance using analytical dashboards. By identifying trends, anomalies, and turning points, supply chain analytics enables timely corrective actions and supports strategic decision-making aimed at sustaining profitability. Graph 3. Month wise analysis

4. Conclusion

The analysis of supply chain cost, selling price, and profit provides critical insights into the financial dynamics of the automobile manufacturing supply chain. The comparative assessment reveals that although the selling price constitutes the largest proportion of revenue, the profit margin remains relatively low due to the dominance of total supply chain cost. This narrow margin underscores the importance of systematic cost control and efficiency enhancement as primary levers for improving profitability.

The cost bifurcation analysis further clarifies that manpower cost is the most significant contributor to total supply chain expenditure, followed by equipment, miscellaneous, transportation, and tool maintenance costs. This cost structure reflects the labor-intensive and capital-intensive nature of automobile manufacturing. The findings suggest that targeted initiatives such as workforce productivity improvement, selective automation, optimized equipment utilization, and logistics rationalization can substantially reduce overall supply chain cost and improve financial performance.

The month-wise analysis highlights considerable variability in profit and cost over time, despite relatively stable selling prices. The observed decline in total cost from December 2022 indicates the positive impact of operational adjustments and cost control measures, whereas the steep drop in profit during earlier periods demonstrates the vulnerability of profitability to demand fluctuations and rising input costs. This divergence confirms that profit performance is highly sensitive to cost behavior rather than pricing alone.

Overall, the results demonstrate that continuous monitoring and analysis of supply chain financial metrics using data analytics dashboards are essential for effective decision-making. By providing timely visibility into cost drivers, profitability trends, and performance deviations, analytics-based supply chain management enables proactive corrective actions and supports long-term financial sustainability in the automobile manufacturing industry.

Conflict of Interest

The authors declare that there is **no conflict of interest** regarding the publication of this paper.

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